

GPU And Cooling System Theory

Slide 1: GPU Identification (Task 1)

- **Brand:** NVIDIA
- **Model Number:** GeForce RTX 3060
- **Type:** Dedicated (Discrete)
- *(Alternative for basic laptops) -> **Brand:** Intel | **Model:** Iris Xe Graphics | **Type:** Integrated*

(Note: Use this example if you have a gaming PC, or check Task Manager > Performance to find your exact model)



Slide 2: Cooling System Components (Task 2)



- **Heatsink (The Metal Block/Fins):** Absorbs extreme heat directly from the processor.
- **Cooling Fan:** Blows ambient air across the heatsink fins to push the trapped heat away from the hardware.
- **Thermal Paste:** (*Sits under the heatsink*) Fills microscopic gaps between the CPU and cooler to maximize heat transfer.

Liquid Cooling System:



Working Process:-

Step 1: The Pump absorbs heat directly from the processor and transfers it into a circulating liquid coolant.

Step 2: The hot liquid is pushed through the sealed tubes and travels up to the Radiator.

Step 3: The radiator disperses the heat across its metal fins, while attached Fans blow that heat out of the PC case.

Step 4: The now-cooled liquid flows back down to the pump block to continuously repeat the cycle.

Slide 3: Air Cooling vs. Liquid Cooling (Task 3)

Feature	Air Cooling Systems	Liquid Cooling (AIO/Custom)
Cost	Highly affordable and budget-friendly.	Expensive; higher upfront cost.
Efficiency (Cooling)	Good for standard use, but can struggle under extreme loads.	Excellent at moving heat quickly; ideal for heavy gaming/workloads.
Maintenance	Low maintenance (just needs occasional dusting).	Requires checking for leaks, pump failures, or liquid top-offs (custom loops).

Slide 4: Gaming & Streaming PC Build (Task 4)

- **Cooling Choice:** Liquid Cooling (AIO)
- **Justification:** Streaming live IPL matches while simultaneously running graphics-heavy games puts maximum stress on both the CPU and GPU. Liquid cooling absorbs and displaces this heat much faster than air, preventing the system from "thermal throttling" (slowing down to survive). Additionally, liquid coolers generally run quieter under heavy loads, ensuring your microphone doesn't pick up loud, distracting fan noises while you stream.